CLINICAL TRIAL FAILURES · DISCONTINUATION RATE · ONCOLOGY PHASE 2,3 · STARTS 2015–2024

PI3K / AKT / mTOR: 10 molecules stopped early, all small molecules

15 stopped trials, but 10 molecules — all small molecules: Copanlisib, Parsaclisib, Taselisib, Vistusertib and 6 more. 13 of the stops were the trial's own verdict and 2 followed a decision taken elsewhere. Against 5.2% for all oncology trials with a resolved drug target and 4.5% across all oncology Phase 2,3 trials in the same window. The rate counts trials that stopped early for efficacy, safety or benefit–risk reasons. Trials that ran to completion and missed their primary endpoint are counted separately below, never in the rate.

8.3%
15 of 181 closed trials (95% CI 5.1%–13.2%)

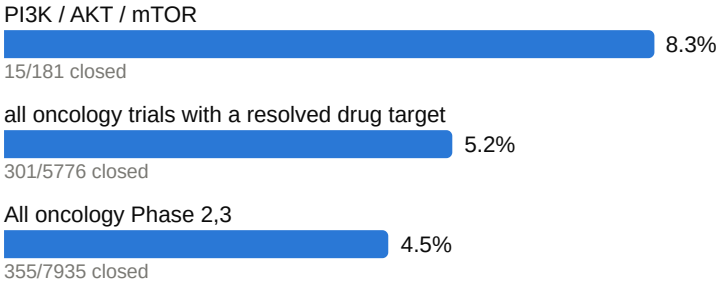
11 / 3 / 1
efficacy / safety / both (adds to 15)

51.3%
of 353 trials have closed · 172 still open or unresolved

2.1%
stopped for a biological reason within 3 years of starting (95% CI 1.0%–4.7%; 202 still at risk; 1.5% area-wide)

5 of 10 **Completed and missed the primary endpoint.** Of 95 completed trials in this segment, 10 posted a primary result on ClinicalTrials.gov that can be read — the sponsor's own comparison held to the sponsor's own threshold, or its statement that the endpoint was missed: 5 missed and 4 met, 1 split across co-primary endpoints. Counted separately and never in the rate above; a floor, not a rate. 32 of the 92 completed trials that finished more than a year ago have posted no results at all — a gap in what can be known, not a sign of failure.

How this compares



What is in this class: 252 distinct experimental drugs across 353 trials (small molecule 352, antibody 93, unknown 18, adc 14). Most tested: ALPELISIB (41), COPANLISIB (38), IPATASERTIB (33), PACLITAXEL (24), FULVESTRANT (24). A class groups drugs by what they act on; check that the grouping is one you would make before reading the rate as a property of the mechanism.

How much does this rest on one decision? The 15 stops came from 12 sponsor-asset programmes across 11 sponsors; the largest (Samsung Medical Center) contributed 2; removing that programme's trials from both sides leaves 7.3%. Screened alongside 85 other segments, the rate is within what screening that many classes could produce (q=1); the molecules and trials that failed here are the record either way. A further 19 closed trials here were terminated with no cause recorded in the registry; if every one of them were biological the rate would be 18.8%. Read it as a screen worth checking against the underlying trials.

Bars show the share of closed trials stopped for biological reasons. Confidence intervals overlap where sample sizes are small — read the counts, not just the bars.

By start cohort

Trials started	Stops	Closed	Rate	95% CI
2015–2017	10	90	11.1%	6.1%–19.3%
2018–2020	3	56	5.4%	1.8%–14.6%
2021–2024	2	35	5.7%	1.6%–18.6%

Where the stops sit

Efficacy / futility: 11, Safety: 3, Efficacy and safety: 1.
Sponsors with most stops: Samsung Medical Center (3), Bayer (2), Incyte (2), Roche (1), Fondazione Italiana Linfomi - ETS (1), National Heart, Lung, and Blood Institute (NHLBI) (1).

The stopped trials

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT02626455	3	Bayer	COPANLISIB	2023-09	Efficacy	Company decided to terminate this study due to study did not meet the primary endpoint.

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT04551053	3	Incyte INCY	PARSACLISIB	2023-08	Efficacy	The study was terminated due to futility.
NCT04551066	3	Incyte INCY	PARSACLISIB	2023-08	Efficacy	The study was terminated due to futility.
NCT03458728	2	Bayer	COPANLISIB	2023-02	Efficacy	no anticipated benefit over available standard therapies
NCT02340221	3	Roche	TASELISIB	2021-06	Efficacy + safety	The Sponsor discontinued the manufacturing and development of taselisib due to modest clinical benefit and limited tolerability.
NCT03890289	2	Fondazione Italiana Linfomi - ETS	IDELALISIB, OBINUTUZUMAB	2021-04	Safety	Study was halted prematurely due to safety, since tocity stopping rules have been met

The 6 most recent of 15 stopped trials. Every molecule behind all 15 is named above; the complete trial list, the trials still running and the terminations with no readable cause ship with the dataset and the diligence report.

Method

Denominator: ClinicalTrials.gov interventional Phase 2,3 oncology trials started 2015–2024 that have closed (completed or terminated). Numerator: terminated trials whose registry stop reason is classified as biological (efficacy, safety or benefit–risk) by Classification V2 — held-out precision 95.5%, recall 95.3% (n=600). Drugs are linked to ChEMBL and the NCI Thesaurus; 76% of industry oncology trials in this window carry a resolved drug target, and a trial without one cannot enter a mechanism class. Intervals are Wilson 95%.

Limits

Not a failure rate: completed trials that missed their endpoints are not in the rate. They are counted separately where the sponsor posted a primary result that can be read — most did not, so that count is a floor — and programmes discontinued after a completed trial do not appear. Stop reasons are sponsor-reported. This is a closed-trial proportion, not a time-to-event analysis: only 51.3% of trials in this segment have closed, and a trial that stops early enters the denominator sooner than one that runs to completion, which can inflate the rate in immature segments. Recent cohorts have fewer closed trials, so their rates are less stable. Research signals, not clinical or investment advice.

The whole cohort, trial by trial. The diligence report names every trial behind these figures — the ones still running and the ones that completed and missed their primary endpoint included — and places your molecule against each one that failed. Free during the beta, in exchange for a work email, delivered immediately: **clinicaltrialfailures.com/asset-check**

Sources: ClinicalTrials.gov (NLM); ChEMBL (EMBL-EBI, CC BY-SA 3.0); NCI Thesaurus (NCI); RxNorm/RxClass (NLM); SEC EDGAR. Classification, linkage and rates by Clinical Trial Failures. Rebuilt weekly; this brief covers trials started 2015–2024.